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Power management integrated circuit with charge pump

Abstract

An apparatus includes a power management integrated circuit (PMIC). The PMIC includes a voltage regulator supplied by a first power source and configured to generate a first output and a charge pump supplied by a second power source and configured to generate a second output. A bias voltage output of the power management integrated circuit is generated based at least in part on the first output and the second output. The charge pump is configured to adjust the second output based at least in part on a comparison between the bias voltage output and a reference voltage.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/527,661 filed on Nov. 16, 2021, which claims the benefit of U.S. Provisional Application No. 63/219,990 filed on Jul. 9, 2021, the entire contents of each of which are incorporated herein by reference.

BACKGROUND OF THE SPECIFICATION

(1) The present disclosure relates to memory. More specifically, the present disclosure relates in some embodiments to voltage regulation in a power management integrated circuit (PMIC) of a double data-rate (DDR) random access memory (RAM) module.

(2) Power consumption and efficiency is increasingly important in both fixed and portable computing devices to reduce power costs and ensure a long battery life. Portable computing devices, such as, e.g., laptops, notebooks, netbooks, or other computing devices that may need to rely on battery power, often have stringent requirements for power consumption and efficiency. In such computing devices, each component typically needs to be optimized to reduce power consumption.

SUMMARY

(3) In an embodiment, an apparatus is disclosed that comprises a power management integrated circuit (PMIC). The PMIC comprises a voltage regulator supplied by a first power source and configured to generate a first output and a charge pump supplied by a second power source and configured to generate a second output. A bias voltage output of the power management integrated circuit is generated based at least in part on the first output and the second output. The charge pump is configured to adjust the second output based at least in part on a comparison between the bias voltage output and a reference voltage.

(4) In another embodiment, a memory module is disclosed that comprises a power management integrated circuit supplied by a first power source and a second power source and configured to

output a bias voltage output. The power management integrated circuit comprises a charge pump supplied by the second power source and configured to generate a charge pump voltage output. The bias voltage output of the power management integrated circuit is generated based at least in part on the charge pump voltage output. The charge pump is configured to adjust the charge pump voltage output based at least in part on a comparison between the bias voltage output and a reference voltage.

(5) In another embodiment, an apparatus is disclosed that comprises a first capacitor that is configured to alternately charge from a power source based at least in part on a first portion of a clock signal and to discharge to a bias voltage output of a power management integrated circuit based at least in part on a second portion of the clock signal. The apparatus further comprises a second capacitor that is configured to alternately charge from the power source based at least in part on the second portion of the clock signal and to discharge to the bias voltage output of the power management integrated circuit based at least in part on the second portion of the clock signal.

(6) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description. In the drawings, like reference numbers indicate identical or functionally similar elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram of an example memory system according to an embodiment of the disclosure.

(2) FIG. 2 is a block diagram illustrating an example memory module of the memory system of FIG. 1 according to an embodiment of the disclosure.

(3) FIG. 3 is a block diagram of an example power management integrated circuit (PMIC) of the memory module of FIG. 2 according to an embodiment of the disclosure.

(4) FIG. 4 is a block diagram of an example charge pump loop of the PMIC of FIG. 3 according to an embodiment of the disclosure.

(5) FIG. 5 is a circuit diagram of a charge pump core of the charge pump loop of FIG. 4 according to an embodiment of the disclosure.

(6) FIG. 6 is a graph of example voltage waveforms output by components of the charge pump core of FIG. 5 according to an embodiment of the disclosure.

(7) FIG. 7 is a circuit diagram of a self-biased switch of the charge pump loop of FIG. 4 according to an embodiment of the disclosure.

DETAILED DESCRIPTION

(8) Power management in DDR random access memory (RAM) modules often relies on the use of a power management integrated circuit (PMIC) to handle the conversion of a bulk power input to one or more power outputs having voltages that correspond to the requirements for different components of the DDR memory module, and in some embodiments, a DDR fifth generation (DDR5) memory module. In one example, the disclosed embodiments may comprise unbuffered dual inline memory modules (UDIMM). For example, for a notebook computer, the disclosed embodiments may comprise small outline dual in-line memory modules (SODIMM) such as, e.g., DDR5 SODIMM. In another example, the disclosed embodiments may comprise registered dual inline memory modules (RDIMM). The disclosed embodiments may alternatively comprise any type of memory module.

(9) FIGS. 1 and 2 illustrate an example embodiment of a memory system 10. The memory system

10 includes memory modules **20.sub.1**, **20.sub.2** . . . **20.sub.N**, also referred to herein collectively or individually as memory module(s) **20**, connectors **70** and a memory controller **80**.

(10) With reference to FIG. **1**, in one example embodiment, the memory modules **20** may comprise dual in-line memory modules (DIMMs). In some embodiments, the memory modules **20** may be implemented as double data rate fifth generation (DDR5) SDRAM modules. While described and illustrated herein as having a particular type, arrangement and number of components, in other embodiments, memory modules **20** may comprise any other type, arrangement or number of components.

(11) An example memory module **20** comprises circuitry blocks **30.sub.1**, **30.sub.2**, **30.sub.3**, **30.sub.4**, **30.sub.5** . . . **30.sub.P-4**, **30.sub.P-3**, **30.sub.P-2**, **30.sub.P-1** and **30.sub.P**, circuitry blocks **40.sub.1**, **40.sub.2**, . . . **40.sub.M-1** and **40.sub.M**, a registered clock driver (RCD) **50**, a PMIC **60**, connectors **70** and any other blocks, circuits, pins, connectors, traces or other component typically found in a memory module. In some embodiments, circuitry blocks **30.sub.1**, **30.sub.2**, **30.sub.3**, **30.sub.4**, **30.sub.5** . . . **30.sub.P-4**, **30.sub.P-3**, **30.sub.P-2**, **30.sub.P-1** and **30.sub.P** may be configured as data buffers and will also be referred to herein collectively or individually as data buffers(s) **30**. In some embodiments, circuitry blocks **40.sub.1**, **40.sub.2**, . . . **40.sub.M-1** and **40.sub.M** may be configured as memory devices and will also be referred to herein collectively or individually as memory device(s) **40**. While described herein as data buffers **30** and memory devices **40**, circuitry blocks **30** and **40** may also or alternatively be utilized for any other purpose by the memory modules **20**.

(12) In some embodiments, data buffers **30** and memory devices **40** comprise synchronous dynamic random-access memory (SDRAM) devices, chips or modules. In some embodiments, data buffers **30** and memory devices **40** also or alternatively comprise any other types of memory devices such as, e.g., SRAM, DRAM, MROM, PROM, EPROM and EEPROM. The data buffers **30**, memory devices **40** or both may be physically located on one side or both sides (e.g., the front and back) of the memory module **20**.

(13) PMIC **60** is configured to perform power management for the memory module **20**. For example, the PMIC **60** may be configured to scale up or scale down voltages, perform DC-DC conversions or perform other similar power management operations. In some embodiments, PMIC **60** may comprise low-dropout regulators (LDOs), DC-DC converters such as, e.g., buck or boost converters, pulse-frequency modulation (PFM), pulse-width modulation (PWM), power field-effect transistors (FETs), real-time clocks (RTCs) or any other circuitry that may typically be found in a PMIC.

(14) Connectors **70** may comprise, for example, pins, traces or other connections that are configured to connect the memory modules **20** to other components of a computing system such as, e.g., a memory controller **80**, motherboard, or other components. In some embodiments, the connectors **70** may comprise, e.g., a 288-pin configuration or any other pin configuration.

(15) In some embodiments, the memory module **20** comprises the connectors **70**. In other embodiments, a motherboard, memory controller **80** or any other component of a computing device comprises the connectors **70**. In another embodiment, one or more of the connectors **70** may be part of the memory module **20** and one or more of the connectors **70** may be part of the motherboard, memory controller **80** or other component of the computing device.

(16) The memory modules **20** may be connected to the motherboard, memory controller **80** or other component of the computing device, e.g., by connectors **70**, to transfer data between components of the computing device and the memory modules **20**. For example, in an embodiment that implements a UDIMM, the connectors **70** may comprise a 64-bit bus, a 72-bit bus or a bus comprising any other number of bits.

(17) The memory modules **20** are shown connected to the memory controller **80** of the computing device. In an example embodiment, the memory controller **80** may be implemented as a component of a computer motherboard, or main board, of the computing device, e.g., on a northbridge of the

motherboard. In another example, the memory controller **80** may be implemented as a component of a microprocessor of the computing device. In yet another example, the memory controller **80** may be implemented as a component of a central processing unit (CPU) of the computing device. In other embodiments, the memory controller **80** may be implemented as a part of any other component of the computing device.

(18) In some embodiments, the memory modules **20** are implemented as DDR5 SDRAM memory modules. As an example, the memory modules **20** may comprise a memory module density of 128 gigabyte (GB), 512 GB, one terabyte (TB), or higher per module. Memory modules **20** may operate with a frequency of about 1.2 to about 3.2 giga-Hertz (GHz) and a data rate range of about 3.2 GT/s to about 4.6 GT/s and in some cases a data rate up to about 8 GT/s or more. In some embodiments, the memory modules **20** may alternatively comprise smaller or larger densities, operate at lower or higher frequencies and operate at lower or higher data rates.

(19) With reference now to FIG. 2, a block diagram illustrating an example memory module **20** of FIG. 1 is shown. The memory module **20** may be representative of the memory modules **20A-20N**. The memory module **20** is shown communicating with the memory controller **80**. The memory controller **80** is shown as part of a circuit **90** such as, e.g., a motherboard, main board or other component of a computing device that communicates with the memory module **20**.

(20) The memory module **20** comprises one or more groupings of circuits **22.sub.1**, **22.sub.2**, **22.sub.3**, **22.sub.4**, **22.sub.5** . . . **22.sub.Q-4**, **22.sub.Q-3**, **22.sub.Q-2**, **22.sub.Q-1** and **22.sub.Q**, also referred to herein collectively or individually as data paths **22** of the memory module **20**. In the example shown, the memory module **20** may comprise five data paths **22**, e.g., data paths **22.sub.1**, **22.sub.2**, **22.sub.3**, **22.sub.4** and **22.sub.5**, on one side of the RCD **50** and five data paths **22**, e.g., data paths **22.sub.Q-4**, **22.sub.Q-3**, **22.sub.Q-2**, **22.sub.Q-1** and **22.sub.Q**, on the other side of the RCD **50**.

(21) In other embodiments, memory module **20** may comprise other arrangements having a greater or smaller number of data paths **22** on each side of the RCD **50**.

(22) The data paths **22** may each comprise a respective memory channel **42.sub.1**, **42.sub.2**, **42.sub.3**, **42.sub.4**, **42.sub.5** . . . **42.sub.R-4**, **42.sub.R-3**, **42.sub.R-2**, **42.sub.R-1** and **42.sub.R**, also referred to herein collectively and individually as memory channel(s) **42**. Each memory channel **42** may comprise one or more of the memory devices **40**. For example, memory channel **42.sub.1** may comprise memory devices **40.sub.1** through **40.sub.S**, while memory channel **42.sub.R** may comprise memory devices **40.sub.T** through **40.sub.M**.

(23) The memory controller **80** is configured to generate a variety of signals including a clock signal (CLK), control signals (ADDR and CMD) and command signals. One or more of the CLK, ADDR and CMD signals may be provided to the RCD **50**, e.g., via one or more buses **23**.

(24) Signals from the memory controller **80** may also be transmitted from the memory controller **80** to the PMIC **60** via a bus **24**, also referred to herein as a host interface bus **24**. In some embodiments, host interface bus **24** is bi-directional and is configured to communicate commands or other data between PMIC **60** and memory controller **80** or other components of the memory module **20**. The host interface bus **24** may implement an I.sup.2C protocol, an I.sup.3C protocol or any other protocol.

(25) A data bus **72** may be connected between the memory controller **80** and the data paths **22**, e.g., with data buffers **30**, and may comprise connectors **70**, e.g., traces, pins and other connections, between the memory controller **80** and the data paths **22**.

(26) The memory controller **80** may generate or receive data signals, e.g., DQa-DQn, and data strobe signals, e.g., DQSa-DQSn, that may be presented to or received from the data bus **72**.

(27) Portions of the signals DQa-DQn and DQSa-DQSn may be presented to or received from respective data paths **22**. In the example shown, each of the signals DQa-DQn may have a corresponding signal DQSa-DQSn. In some embodiments, one DQS signal may strobe multiple DQ signals, e.g., one DQS signal for four DQ signals in some embodiments.

(28) The RCD **50** is configured to communicate with the memory controller **80**, the data buffers **30**, the memory channels **42** and the PMIC **60**. The RCD **50** is configured to decode instructions, e.g., control words, received from the memory controller **80**. For example, the RCD **50** may be configured to receive and decode register command words (RCWs). In another example, the RCD **50** may be configured to receive and decode buffer control words (BCWs). The RCD **50** is configured to train one or more of the data buffers **30**, memory devices **40** and the command and address lines between the RCD **50** and the memory controller **80**. For example, the RCWs may flow from the memory controller **80** to the RCD **50** and be used to configure the RCD **50**.

(29) In some embodiments, the RCD **50** may implement a command/address register, e.g., a 32-bit 1:2 command/address register. The RCD **50** may support an at-speed bus, e.g., a unidirectional buffer communications (BCOM) bus between the RCD **50** and the data buffers **30**. In some embodiments, the RCD **50** may implement one or more of automatic impedance calibration, command/address parity checking, control register RCW readback, a serial bus such as, e.g., a 1 MHz inter-integrated circuit (I^{sup}.2C) bus, and a 12.5 MHz inter-integrated circuit (PC) bus. Inputs to the RCD **50** may be pseudo-differential using one or more of external and internal voltages. The clock outputs, command/address outputs, control outputs and data buffer control outputs of the RCD **50** may be enabled in groups and independently driven with different strengths.

(30) The RCD **50** is configured to receive the CLK, ADDR and CMD signals or other signals such as, e.g., RCWs and BCWs, from the memory controller **80** and to utilize various digital logic components to generate corresponding output signals based on the CLK, ADDR and CMD signals. For example, the RCD **50** is configured to generate corresponding signals such as, e.g., CLK', ADDR' and CMD' signals based on the received CLK, ADDR and CMD signals. The CLK', ADDR' and CMD' signals may be presented to the memory channels **42**. For example, the CLK' signals may be transmitted from the RCD **50** to the memory channels **42** on a common bus **25** and the ADDR' and CMD' signals may be transmitted from the RCD **50** to the memory channels **42** on a common bus **26**. The RCD **50** is also configured to generate one or more data buffer control (DBC) signals that are transmitted to the data buffers **30**, for example, on a common bus **27**, also referred to herein as a data buffer control bus **27**.

(31) The data buffers **30** are configured to receive commands and data from the data buffer control bus **27** and to generate data, receive data or transmit data to and from the data bus **72**. Each data path **22** also comprises bus **28** between its data buffer **30** and memory channel **42** that is configured to carry the data between the data buffer **30** and memory channel **42**. For example, as seen in FIG. **2**, data path **22.sub.1** comprises a bus **28** between data buffer **30.sub.1** and memory channel **42.sub.1**.

(32) The data buffers **30** are configured to buffer data on the buses **72** and **28** for write operations, e.g., data transfers from the memory controller **80** to the corresponding memory channels **42**, and read operations, e.g., data transfers from the corresponding memory channels **42** to the memory controller **80**.

(33) In some example embodiments, the data buffers **30** exchange data with the memory devices **40** via the buses **28** in small units, e.g., 4-bit nibbles. In other embodiments, larger or smaller sizes of data transfer may alternatively be utilized. In some cases, the memory devices **40** may be arranged into multiple sets, e.g., two sets. For example, for a two set/two memory device implementation, e.g., memory devices **40.sub.1** and **40.sub.2**, each set may contain a single memory device **40**, e.g., **40.sub.1** or **40.sub.2**) with each memory device **40** being connected to the respective data buffers **30** through an upper nibble and a lower nibble. For two set/four memory device implementation, each set may contain two memory devices **40**. The first set may be connected to the respective data buffers **30** through the upper nibble and the second set may be connected to the respective data buffers **30** through the lower nibble. For two set/eight memory device implementation, each set may contain four of the memory devices **40**. The first set of four memory devices **40** may connect to the respective data buffers **30** through the upper nibble and the second set of four memory

devices may connect to the respective data buffers **30** through the lower nibble. Other numbers of sets, other numbers of memory devices per set and other data unit sizes may alternatively be utilized.

(34) Memory module **20** may also comprise an interface **29** that is configured to enable communication between the RCD **50** and the PMIC **60**. For example, the interface **29** may be utilized as part of a register clock driver/power management integrated circuit interface, e.g., an RCD-PMIC interface. The interface **29** is configured to support one or more signals or connections that may be bidirectional or unidirectional.

(35) Illustrative embodiments will now be described with reference to FIGS. **1-7**.

(36) Power management in a DDR memory module is performed by a highly integrated, highly efficient PMIC **60** with programmable and precise output voltage power solutions. In some embodiments, the PMIC **60** is powered from two input power supplies, e.g., a management power supply VIN.sub.MGMT and a bulk power supply VIN.sub.BULK although other power supplies may also be present. In some embodiments, PMIC **60** may comprise a plurality of pins, e.g., **36** pins or another number of pins, that connect to the memory module **20** where VIN.sub.MGMT and VIN.sub.BULK each receive power from one or more of the pins. For example, VIN.sub.BULK may receive power from the memory module **20** via four pins and VIN.sub.MGMT may receive power from the memory module **20** via one pin. In other embodiments, a different number of pins may supply power to VIN.sub.MGMT and VIN.sub.BULK. One or more of the pins may correspond to one or more of the connectors **70** where, for example, the power supplied to the PMIC **60** for one or both of VIN.sub.MGMT and VIN.sub.BULK may be received from the circuit **90** of the computing device such as, e.g., a motherboard, main board or other component of the computing device that communicates with the memory module **20**.

(37) In some embodiments, PMIC **60** receives a VIN.sub.MGMT having a voltage of about 3.3V and receives a VIN.sub.BULK having a voltage range of about 4.25V to about 15V from the memory module **20**. For example, the voltage of VIN.sub.BULK may vary based on the required load for the PMIC **60** at any given time. Other voltage values or ranges for VIN.sub.MGMT and VIN.sub.BULK may alternatively be used.

(38) PMIC **60** comprises a plurality of voltage regulation modules (VRMs) **102** through **116**. The VRMs **102** through **116** are configured to provide regulated output voltages for the various components of the memory module **20**. The PMIC **60** is configured to manage, maintain and adjust the output voltages where, for example, the PMIC **60** may perform adjustments or modifications to the output voltages based on instructions received from the RCD **50** or the memory controller **80**.

(39) As illustrated in FIG. **3**, for example, VRMs **102** through **108** comprise DC-DC converters such as, e.g., buck converters, which are utilized to step down the voltage of VIN.sub.BULK to corresponding output voltages VOUT.sub.BUCK A, VOUT.sub.BUCK B, VOUT.sub.BUCK C and VOUT.sub.BUCK D. For example, VRM **102** may output a stepped down output voltage VOUT.sub.BUCK A of about 1.0V, sometimes referred to as a V.sub.DD supply rail, VRM **104** may output a stepped down output voltage VOUT.sub.BUCK B of about 1.0V, also sometimes referred to as a V.sub.DD supply rail, VRM **106** may output a stepped down output voltage VOUT.sub.BUCK C of about 1.1V, sometimes referred to as a V.sub.DDQ supply rail, and VRM **108** may output a stepped down output voltage VOUT.sub.BUCK D of about 1.8V, sometimes referred to as a V.sub.PP supply rail. VRMs **102** through **108** may also be referred to herein as bucks A through D. While described and illustrated as buck converters, VRMs **102** through **108** may alternatively comprise other types of VRMs. In addition, while described above as outputting particular voltage values, VRMs **102** through **108** may alternatively be configured to output other voltage values, for example, as required by the various components of the memory module **20**.

(40) VRMs **110** through **116** comprise LDOs that are configured to convert the voltages of VIN.sub.BULK and VIN.sub.MGMT into continuously controlled, steady, low-noise DC output voltages. For example, VRM **110** is configured to convert VIN.sub.BULK into an output voltage

VOUT.sub.1 of about 1.8V, VRM **112** is configured to convert VIN.sub.MGMT into the output voltage VOUT.sub.1, VRM **114** is downstream of VRMs **110** and **112** and is configured to convert VOUT.sub.1 into an output voltage VOUT.sub.2 of about 1.0V and VRM **116** is configured to convert VIN.sub.BULK into an output voltage V.sub.BIAS of about 5.0V. As seen in FIG. 3, VRMs **110** and **112** work together to maintain or output the voltage VOUT.sub.1 and VRM **116** works together with a charge pump loop **118** to maintain or output the voltage V.sub.BIAS.

(41) While VRMs **102** through **116** are described above as comprising components such as, e.g., buck converters and LDOs, in other embodiments alternative circuitry may be utilized to perform the voltage regulation functions. Similarly, while VRMs **102** through **116** are described above as outputting particular voltages, in other embodiments alternative voltages may be output depending on the requirements of the memory module **20**.

(42) Charge pump loop **118** comprises, for example, a switched capacitor voltage converter or another converter that is configured to adjust its output voltage to maintain V.sub.BIAS at or about a target voltage, e.g., 5V. For example, the outputs of VRM **116** and charge pump loop **118** may combine and compensate each other to help PMIC **60** save power and gain efficiency benefits. In some embodiments, which one of VRM **116** and charge pump loop **118** outputs V.sub.BIAS may be switched depending on which one provides the best conversion efficiency.

(43) With reference now to FIG. 4, an example implementation of charge pump loop **118** is illustrated. In the charge pump loop **118**, an error amplifier **120** receives a reference voltage V.sub.REF and an output V.sub.FEEDBACK from a feedback circuit **122** as inputs. The feedback circuit **122** comprises a resistor divider or other component that adjusts the V.sub.BIAS voltage for use by the error amplifier **120**. For example, the feedback circuit **122** may divide V.sub.BIAS by a constant value N ($V_{sub.BIAS}/N$) to output an adjusted voltage V.sub.FEEDBACK which is used as an input for error amplifier **120**. In other embodiments, other values or mathematical operations may be performed on V.sub.BIAS by feedback circuit **122** to generate V.sub.FEEDBACK.

(44) In an example embodiment, V.sub.FEEDBACK is compared to V.sub.REF by error amplifier **120**. When V.sub.FEEDBACK has a smaller voltage than V.sub.REF, the error amplifier **120** will drive a voltage-controlled oscillator (VCO) **124** to output a higher frequency value. For example, as the difference between V.sub.FEEDBACK and V.sub.REF increases, the frequency output by VCO **124** will also increase. Increasing the frequency output by VCO **124** drives a charge pump core **126** to deliver more charge to an on-board-capacitor CP.sub.BOARD (FIG. 5) via a self-bias switch (SBS) **128** in addition to supplying a load on V.sub.BIAS and increasing V.sub.BIAS.

(45) When V.sub.FEEDBACK has a larger voltage than V.sub.REF, the error amplifier **120** will drive VCO **124** to output a lower frequency value. For example, as the difference between V.sub.FEEDBACK and V.sub.REF increases, the frequency output by VCO **124** will decrease. A lower frequency output by VCO **124** drives the charge pump core **126** to deliver less charge to CP.sub.BOARD via SBS **128** in addition to supplying a load on V.sub.BIAS and decreasing V.sub.BIAS.

(46) When V.sub.FEEDBACK is equal to V.sub.REF, VCO **124** outputs a stable frequency and V.sub.BIAS is settled at $V_{sub.REF} * N$. For example, V.sub.BIAS may settle at 5V in some embodiments.

(47) With reference now to FIG. 5, the charge pump core **126** is illustrated in more detail. In some embodiments, charge pump core **126** comprises a switched capacitor voltage converter having two sides, a left side and a right side. Each side of the charge pump core **126** has two phases, a charge phase and a pump phase, and comprises a variety of components including n channel transistors Nch.sub.CHARGE and Nch.sub.GND, e.g., high side FETs, p channel transistors Pch.sub.DRV and Pch.sub.TO_LOAD, e.g., low side FETs, and a capacitor CP.sub.FLT.

(48) During the charge phase for a given side, the capacitor CP.sub.FLT is connected between VIN.sub.MGMT or and ground (GND) through transistors Nch.sub.CHARGE and Nch.sub.GND. This allows the capacitor CP.sub.FLT to be charged to the voltage of VIN.sub.MGMT, e.g., 3.3V.

(49) During the pump phase for a given side, CP.sub.FLT is connected between VIN.sub.MGMT and Netv.sub.cp through transistors Pch.sub.DRV and Pch.sub.TO_LOAD to discharge the capacitor CP.sub.FLT. The amount of charge, $(2V_{IN.sub.MGMT} - V_{sub.BIAS}) * CP_{sub.FLT}$, is pumped to the capacitor CP.sub.BOARD and drained by the load during a half clock cycle to maintain the voltage of V.sub.BIAS at around the target value, e.g., 5V in this embodiment as shown by the waveform in FIG. 6 for Netv.sub.cp.

(50) The left and right sides of the charge pump core **126** take turns performing the charge and pump phases. For example, as can be seen in FIG. 5, when the left side of the charge pump core **126** is charging, Net.sub.XP is set equal to VIN.sub.MGMT which turns off the transistor NCh.sub.CHARGE of the right side but also turns on the transistor Pch.sub.TO_LOAD of the right side to allow the right side to discharge its capacitor CP.sub.FLT to Netv.sub.cp and settle the V.sub.BIAS, e.g., at 5V. Similarly, when the right side of the charge pump core **126** is charging, Net.sub.XN is set equal to VIN.sub.MGMT which turns off the transistor Nch.sub.CHARGE of the left side but also turns on the transistor Pch.sub.TO_LOAD of the left side to allow the left side to discharge its capacitor CP.sub.FLT to Netv.sub.cp and settle the V.sub.BIAS, e.g., to 5 V.

(51) The switching between charge and pump phases for the sides of the charge pump core **126** are controlled by the output frequency of the VCO **124**. For example, as the frequency output by VCO **124** increases, the rate of switching between charging and discharging for the left and right sides also increases. Likewise, as the frequency output by VCO **124** decreases, the rate of switching between charging and discharging for the left and right sides decreases. In illustrative embodiments, the capacitor CP.sub.FLT on each side of the charge pump core **126** is fully charged or discharges during each phase. In other embodiments, for example, the frequency output by VCO **124** may be greater than a threshold value at which the CP.sub.FLT may not fully charge or discharge during each phase, e.g., the frequency is so high that the phases switch before a full charge or discharge of the CP.sub.FLT can occur. In some embodiments, the frequency output range of the VCO **124** may be controlled such that the capacitors CP.sub.FLT will fully discharge or charge before switching phases, e.g., by inhibiting the maximum frequency output by VCO **124** from limiting the maximum output current of the charge pump core **126**.

(52) The charging cycles of the charge pump core **126** may follow equations (1) and (2) below:

$$(53) \quad V_{BIAS} = V_{REF} * N \quad (1) \quad (2V_{IN_{MGMT}} - V_{BIAS}) * CP_{FLT} = I_{load} * T * 0.5 \quad (2)$$

(54) N is a coefficient that may be determined and set based on the target voltage for V.sub.BIAS and the value of the reference voltage V.sub.REF. For example, if the target voltage for V.sub.BIAS is 5V and the reference voltage V.sub.REF is 1.8V, N is equal to about 2.777. In other embodiments, if V.sub.REF is equal to 1V, N may be equal to 5. Any other value for N may be utilized based on the target voltage for V.sub.BIAS and the reference voltage V.sub.REF.

(55) From VIN.sub.MGMT to V.sub.BIAS, the theory efficiency of charge pump loop **118** is about 76%, e.g., $V_{sub.BIAS}/VIN_{sub.MGMT}/2$. From VIN.sub.BULK to V.sub.BIAS, the theory efficiency of VRM **116**, e.g., an LDO, is about 42%, e.g., $V_{sub.BIAS}/VIN_{sub.BULK}$, assuming VIN.sub.BULK is equal to 12V. By utilizing a charge pump to complement and assist the VRM **116** in maintaining V.sub.BIAS at the target voltage, e.g., 5V, greater efficiency in the PMIC **60** may be achieved over just using an LDO or other VRM to output V.sub.BIAS. This is because the efficiency to step down the voltage of a 12V VIN.sub.BULK power supply to 5V may be much worse than the efficiency to step up the voltage of the 3.3V VIN.sub.MGMT power supply to 5V.

(56) As an example, if the load on the memory module **20** is light, VIN.sub.BULK may be set to 5V such that maintaining the V.sub.BIAS using the VRM **116** may be highly efficient, e.g., have a high theory efficiency. In such a case, the charge pump core **126** may be disabled, e.g., by turning off the SBS **128**. However, as the load on the memory module **20** increases, the voltage of VIN.sub.BULK will also increase which results in greater efficiency losses in the VRM **116** when converting from VIN.sub.BULK and V.sub.BIAS. For example, where little to no efficiency loss

may occur when the VIN.sub.BULK has a voltage of 5V, when VIN.sub.BULK increases to 12V due to increased load requirements, the efficiency loss converting from 12V to 5V using the VRM **116** may be significant, e.g., a theory efficiency of 42% as mentioned above. To improve the efficiency in maintaining V.sub.BIAS at the target voltage in such a case, VRM **116** may be turned off and charge pump core **126** may be turned on, e.g., by turning on SBS **128**. In some embodiments, both the VRM **116** and the charge pump core **126** may be turned on at the same time where, for example, the charge pump core **126** may assist the VRM **116** in maintaining the target voltage of V.sub.BIAS thereby reducing the efficiency loss in the conversion from VIN.sub.BULK to V.sub.BIAS.

(57) With reference to FIG. **6**, waveforms for the voltage nodes Net.sub.XP, Net.sub.CK_N, Net.sub.XN, Net.sub.CK_P and Netv.sub.cp of the charge pump core **126** of FIG. **5** during the pump and charge phases are illustrated. When the left side of charge pump core **126** is on a pump phase, Net.sub.XP increases its voltage due to the discharge of the left side capacitor CP.sub.FLT and then dumps the charge to the capacitor CP.sub.BOARD before settling at the voltage of V.sub.BIAS, e.g., 5V. The voltage node Net.sub.CK_N is connected to VIN.sub.MGMT through the left side Pch.sub.DRV and helps Net.sub.XP increase by providing potential to the back side of the left side CP.sub.FLT and then settles at the voltage of VIN.sub.MGMT, e.g., 3.3V.

(58) At the same time, the right side of charge pump core **126** is on the charge phase. Net.sub.XN is connected to VIN.sub.MGMT through the right side NCh.sub.CHARGE and Net.sub.CK_P is connected to the GND through right side Nch.sub.GND. The right side CP.sub.FLT is charged to VIN.sub.MGMT.

(59) Since the clock frequency output by VCO **124** drives the charge phase and pump phase switching, the left side of the charge pump core **126** will be on the charge phase during the next part of the clock cycle and the right side of the charge pump core **126** will be on pump phase. As can be seen in FIG. **6**, the waveform for Netv.sub.cp shows a small ripple on V.sub.BIAS around the target of 5V, e.g., fluctuating near 5V, which is related to the ratio of CP.sub.FLT/CP.sub.BOARD.

(60) With reference now to FIG. **7**, the SBS **128** combines the output Netv.sub.cp of charge pump core **126** with the output of VRM **116** to generate V.sub.BIAS. Pch.sub.SW is a transistor switch that connects Netv.sub.cp with the stepped down output of VRM **116** from VIN.sub.BULK. The nwell of Pch.sub.SW is biased by transistors Pch.sub.B1 or Pch.sub.B2. For example, when Pch.sub.SW is turned on, both Pch.sub.B1 and Pch.sub.B2 are turned off and the nwell of Pch.sub.SW is biased through parasitic diodes D1 and D2. The SBS **128** inhibits the parasitic path of the diodes D1 and D2 between Netv.sub.cp and the output of VRM **116**. For example, when VRM **116** is turned off, its output has a high impedance and V.sub.BIAS is driven by Netv.sub.cp. When the charge pump core **126** is turned off, e.g., by turning off Pch.sub.SW, the output of SBS **128** has a high impedance with V.sub.BIAS and head-to-head diodes D1 and D2 inhibit the parasitic current path from V.sub.BIAS to VIN.sub.MGMT, e.g., via charge pump core **126**.

(61) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(62) The corresponding structures, materials, acts, and equivalents of all means or step plus function elements, if any, in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The disclosed embodiments of the present invention have been presented for purposes of illustration and description but are not intended to be exhaustive or limited to the invention in the

forms disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the invention. The embodiments were chosen and described in order to best explain the principles of the invention and the practical application, and to enable others of ordinary skill in the art to understand the invention for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A power management circuit for supplying a power to a load, the power management circuit comprising: a voltage regulator configured to receive a first power source, generate a first output, and be controlled to be on state or off state; a charge pump configured to receive a second power source, generate a second output, and be controlled to be on state or off state, wherein a bias voltage output of the power management circuit is generated based at least in part on the first output and the second output, wherein the on states and off states of the voltage regulator and the charge pump are controlled based on a voltage value of the first power source, and wherein the voltage regulator and the charge pump are coupled in parallel to the bias voltage output respectively, and the first power source and second power source are independent power sources and provided to the voltage regulator and the charge pump respectively; and a switch disposed between the output of the charge pump and the output of the voltage regulator, the switch being configured to switch between outputting the second output and having a high output impedance.
 2. The power management circuit according to claim 1, wherein the second power source has a voltage that is smaller than a voltage of the first power source.
 3. The power management circuit according to claim 1, wherein the first power source is a variable voltage power source and the second power source is a static voltage power source.
 4. The power management circuit according to claim 1, wherein when the voltage value of the first power source increases, the charge pump is controlled to be the on state.
 5. The power management circuit according to claim 1, wherein when the voltage value of the first power source increases, the voltage regulator is controlled to be the off state.
 6. The power management circuit according to claim 1, wherein the charge pump further comprises a switch, and the charge pump is controlled to be the off state by turning off the switch.
 7. The power management circuit according to claim 1, wherein the load is memory DDR random access memory module.
 8. The power management circuit according to claim 7, wherein the first power source is bulk power input and the second power source is management power supply.
 9. The power management circuit according to claim 7, wherein the on states and off states of the voltage regulator and the charge pump are controlled by a memory controller or a registered clock driver.
 10. The power management circuit according to claim 1, wherein the switch comprises a self-biased switch.
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